

Cryptography Basics

Cryptography is the science of securing information by transforming it into a form that is unreadable to unauthorized parties, while still allowing intended recipients to access the original content. At its core, cryptography relies on algorithms that convert plaintext, the original readable data, into ciphertext, an encoded version that appears as meaningless data to anyone without the proper key. There are two primary types of encryption used in modern cryptography: symmetric encryption and asymmetric encryption. Symmetric encryption uses the same key for both encrypting and decrypting data, making it fast and efficient, but requiring that the key be securely shared between parties beforehand. Common symmetric encryption algorithms include the Advanced Encryption Standard, widely used to protect sensitive data due to its strong security and efficiency. Asymmetric encryption, also known as public key cryptography, uses a pair of mathematically related keys, a public key that can be shared openly and a private key that must be kept secret, allowing secure communication without the need to exchange a shared secret in advance. This approach underpins many secure communication protocols used across the internet today. Hash functions represent another important cryptographic tool, converting data of any size into a fixed-length string of characters, a process that is designed to be irreversible and highly sensitive to even small changes in the input data. Hash functions are widely used for verifying data integrity and securely storing passwords, since even providers themselves cannot reverse a properly hashed password back to its original form. As computing power increases, cryptographic algorithms must continually evolve to resist new forms of attack, including emerging threats posed by advances in quantum computing.